

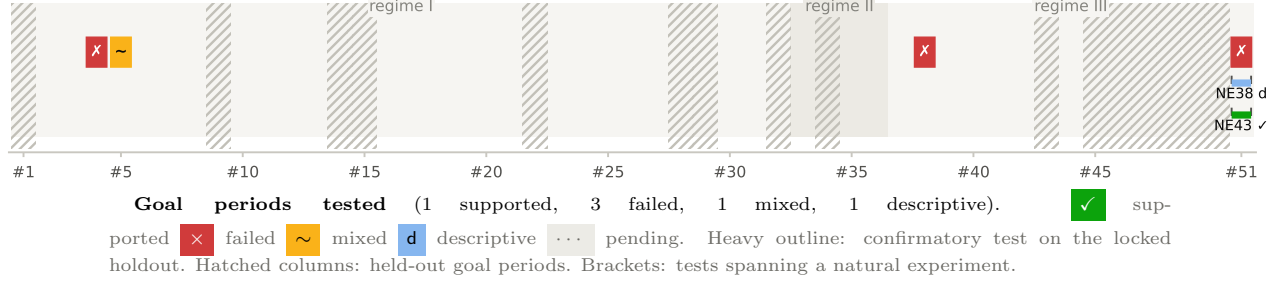
H59: One lever model for all operator inputs

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

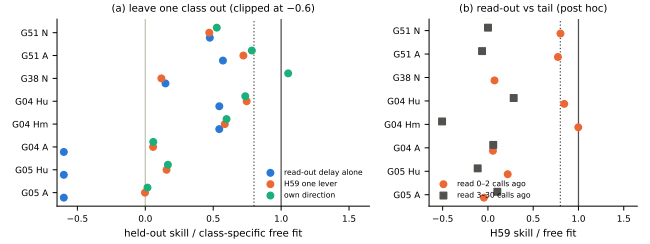
Bottom line. One triple per input class does not predict a held-out class. Only the response at the read-out call is shared; the classes differ in their tails and, in some regimes, in which transition they act on.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Can every operator input be summarized by one triple: field strength h , catalytic strength κ and read-out delay d ? If so, a model fitted on three input classes should predict the fourth. Each model call was coded as idle, work or talk. Every nudge, broadcast human message, named human message and @-mention was timed at the call that first read it (DQ1 ledger). One generalized linear response with a shared kernel and a shared field direction was fitted per period, and each class in turn was held out (5 day folds). Synthetic validation came first. Replication: G51, G38, G04; natives: NE43, #5 (G05), NE38. **The claim fails.** Held-out-class skill is 0.47 (nudges) and 0.72 (mentions) of a class-specific fit in G51, 0.06–0.75 in G04, 0.12 for G38 nudges. What holds is narrower: at the read-out itself (0–2 calls) the shared shape keeps 77–100% of the skill in G51 and for G04 humans. The classes differ in their tails. Holdout not run.



Observables. (a) Held-out skill of each rival relative to the class-specific free fit, per period and class (informative cells only). H59 (orange) is below the 0.8 line everywhere; the read-out-delay rival (blue) is far below for message classes. (b) Post hoc split: at the read-out (rows read 0–2 calls ago) H59 transfers for G51 nudges and mentions and G04 human messages; in the tail (3–30 calls) it does not.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful. At the moment an agent reads it, an operator input does nearly the same thing whatever its class: the agent is pulled toward talking, with a little extra traffic. A nudge and an @-mention look alike at the receiving call. So one number per class plus the delay until the next model call describes most of the immediate effect. What differs is what follows. Nudged agents stay idle-prone for 30 calls, which looks like the nudger choosing stuck agents rather than a lever effect. Regime-I mentions speed up the work↔wait edge. So for planning, treat an input's immediate pull as portable and its aftermath as class-specific. Reading several messages at once counts as one read (G05), and the class levers do not change when the nudger is switched off (NE43).

PHYSICAL MODEL

Models 02 (kinetic response; field vs. catalyst split of a rate change, as in H39) and 09 (exogenous kicks with a response kernel, on the call clock). Transition log-odds against staying get a baseline (agent, day, aging, Markov-2) plus a kick term. H59 constrains each class to a uniform catalyst κ_c plus a field h_c along one shared direction θ , with one shared kernel K_ℓ over lags since the receiving call. Rivals: class-specific free responses (36 numbers per class), a class-specific direction, the read-out delay alone (one amplitude for all) and the inert model.

$$\eta_{ij} = \alpha_{ij} \cdot x + \sum_{c,\ell} D_{c\ell} \beta_{ij}(c, \ell)$$

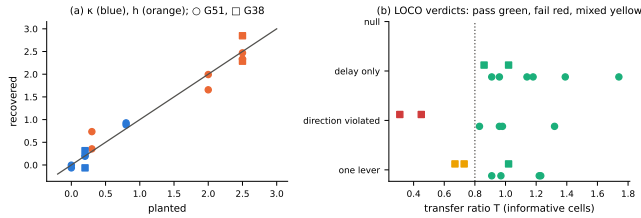
$$\text{H59: } \beta_{ij}(c, \ell) = K_\ell \left[\kappa_c + \frac{h_c}{2} (u_j - u_i) \right], \quad u = (0, \cos \theta, \sin \theta)$$

$$T_c = S_{\text{H59}} / S_{\text{free}}, \quad S = \text{held-out log-lik gain over inert}$$

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RESULTS VS PREDICTIONS

Prediction	Observed	
P1 LOCO $T \geq 0.8$ (G51)	nudges 0.47, mentions 0.72	×
P2 delay alone loses	messages yes; nudges tie	~
P3 kernel at read-out	K_1 0.44 (G51), 0.65 (G38)	~
P4 no inbox decay	nudges, mentions yes; broadcast humans 0.57	~
P5 lever-table signs	all but broadcast-human field size	✓
P6 G38, G04	both fail	×
P7 synthetic	recovery yes; direction blind at idle	~
NE43 invariance	Δ CIs $\ni 0$; transfer 1.01	✓
G05 dose saturates	≥ 3 vs 1 msg: -0.02 ; LOCO fails	~



Second figure. Synthetic validation with the real exposure design. (a) κ and h are recovered at G51 and G38 counts. (b) LOCO verdicts by planted world: a one-lever world passes at G51. A direction violation is caught at G38 but not at G51, because nudges are read almost only by idle agents, where two parameters fit any direction.

GOAL PERIODS IN DETAIL

G04 × failed
G05 ~ mixed
G38 × failed
G51 × failed
NE38 d descriptive
NE43 ✓ supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The read-out/tail split is post hoc (frozen for the holdout as C1). The state space is behavioral only; content fields are not in it. The baseline is fitted once and held fixed in the bootstraps. Named-human (92 reads) and G38 broadcast-human cells are uninformative. Kickoffs are not testable at call level. The nudge tail may be the nudger's targeting, which the past-only baseline cannot see.

ROUND 2

Run `confirm.py` on the #51 tail. Test whether the nudge tail is selection (trigger state, first vs re-fire). Try a two-component lever (edge catalyst + second direction). Add content drift on the same clock. Pool kickoffs across regime-III periods.

- **Essence.** a compact, portable description of what an operator input does. Round 1 says the portable part is the read-out response, and what differs between classes is their tails and, in some regimes, which edge they act on.
- **H59-R1** Run `confirm.py` on the #51 tail after commit and disclosure.
- **H59-R2** Separate tail from selection.
- **H59-R3** Two-component lever.
- **H59-R4** Content layer.
- **H59-R5** Kickoffs across regime-III periods with partial pooling (exception d), as the fifth class.

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